

2016.0 RANGE ROVER (LG), 415-01

INFORMATION AND ENTERTAINMENT SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description and operation of the information and entertainment system, refer to the relevant description and operation section of the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- All diagnostic equipment should comply with local legislation.
- Relevant diagnostic equipment should be regularly checked and calibrated according to the manufacturer's instructions.
- The workshop should be equipped with a full range of general equipment which is to be kept in good order and available to all suitably trained staff.
- Diagnostic equipment must meet the JLR Minimum Standards for general equipment as outlined in TOPIx.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Check all audio system modules ▪ Compact disc player jammed, not loading ▪ Scratched/dirty compact discs	▪ Fuses ▪ Electrical harnesses ▪ Harness connectors

▪ Speakers ▪ Switch(s) stuck or damaged ▪ Loose items in door pockets or glove box rattling	▪ Battery condition, state of charge
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3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SPEAKER DIAGNOSTICS

The symptom chart below should be used when diagnosing speaker faults and must be worked through prior to replacing any speakers

REPORTED SYMPTOM	SYMPTOM DESCRIPTION	POTENTIAL CAUSES (FOR GUIDANCE ONLY)	RECOMMENDED ACTION
▪ Speaker Buzz	▪ Periodic high frequency sound (sounds like an insect buzzing)	▪ Resonance in a trim component, e.g. NVH (noise, vibration and harshness) insulating cloth against speaker grill mesh ▪ Lack of retention of interior and exterior trim panels ▪ Loose harness/harness clips ▪ Labels vibrating against trim/nearby components ▪ Nearby modules	▪ 1. Replicate fault and confirm audio source(s) affected: DAB/CD/Bluetooth via phone/USB via phone/USB via iPod®/AM/FM radio ▪ 2. If fault is specific to one source, audio source must be investigated further as a speaker failure in this instance is less likely, e.g. if issue is only seen on DAB, check DAB module is fully functioning correctly and software is up to date

		vibrating against trim/BIW ▪ Mechanical failure of internal speaker component	▪ 3. Confirm if issue is heard on stereo and/or surround sound settings ▪ 4. Isolate the customer symptom to specific vehicle area using balance and fade audio settings. Check for any loose objects in the area, e.g. pens, keys, coins, etc ▪ 5. Check for any previous work carried out in this area by referring to DDW to help in diagnosing the issue ▪ 6. Check SDD/Topix for any audio related SSM/TSB and software updates. Carry out service updates as required. Retest to confirm if issue is resolved ▪ 7. Apply pressure by hand to door/trim panel and nearby modules (i.e. switchpacks) to confirm if there is a change in symptom sound ▪ 8. If issue is still present, or condition changes by applying pressure to the trim, remove internal trim and apply gentle pressure to components within the door to see if sound changes. This will also allow access to relevant speaker ▪ 9. If symptom condition is unchanged by applying pressure to the trim, examine trim to find source of audio issue and fix as appropriate using Squeaks and Rattle kit LTB00389. If audio issue is still present then move to step 10 ▪ 10. Check around front face of speaker for any
▪ Speaker Rattle	▪ Sounds like loose components rattling around in / near the speaker	▪ Loose trim ▪ Loose items in or around speaker ▪ Loose fixings ▪ Debris inside speaker ▪ Loose harness	
▪ Speaker Hiss / Static	▪ Wideband noise (White Noise) / interference (such as experienced from a poorly tuned radio signal)	▪ Loose connection at speaker ▪ Loose connection at amplifier ▪ Loose connection at audio head unit ▪ Non JLR-approved equipment installed (e.g. USB cables) ▪ Damage to harness connected to the speaker ▪ Audio amplifier fault ▪ Audio head unit fault	
▪ Speaker Crackle	▪ Electrical crackling noise (such as from a loose electrical connection) ▪ A rapid succession of short sharp noises ▪ Electrical interference	▪ Loose connection at speaker ▪ Loose connection at amplifier ▪ Loose connection at audio head unit ▪ Non JLR-approved equipment installed (e.g. USB cables) ▪ Damage to harness connected to the speaker ▪ Audio amplifier fault	

		▪ Audio head unit fault ▪ Internal electrical issue in speaker	loose items that may be touching the speaker, e.g. debris, loose fixings, etc
▪ Speaker Distorted	▪ No significant extraneous noise, but audio reproduction is not as expected e.g. is not clean sounding	▪ Loose trim ▪ Non JLR-approved equipment installed (e.g. USB cables) ▪ Resonance in a trim component, e.g. NVH (noise, vibration and harshness) insulating cloth against speaker grill mesh ▪ Lack of retention of interior and exterior trim panels, nearby harnesses/components ▪ Mechanical failure of internal speaker component	▪ 11. Check all speaker fixing screws are secure and correctly torqued, refer to workshop manual for correct torque figures ▪ 12. Check harness assembly is not vibrating against speaker unit or nearby trim/components and that it is clipped and routed correctly ▪ 13. Check harness assembly is not trapped or impeding the speaker unit and rectify as required ▪ 14. Check harness assembly connections are fully inserted and secured to speaker and amplifier. Also check harnesses are securely connected to nearby components ▪ 15. Check for harness damage and repair/replace as required ▪ 16. If a gasket is present on speaker, without removing the speaker visually check it is correctly seated against speaker surface and not noticeably damaged ▪ 17. If audio issue is still present, remove speaker from mountings, retest speaker in hand to see if the fault is still present ▪ 18. Check for debris in speaker and remove. Retest speaker in hand ▪ 19. Check for damage to speaker and replace speaker if damage is present

			20. Connect new speaker to harness, check new speaker in hand to ensure fault has been rectified. If issue has been resolved, reassemble trim with new trim retention clips 21. Check if audio issue is still present. If issue has been resolved, reassemble trim and retest. If issue is still present, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual
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PINPOINT TESTS FOR SUSPECTED SPEAKER FAULTS

NOTE:

See separate Pinpoint Tests (below) for sub-woofer faults

PINPOINT TEST A : NO SOUND OUTPUT FROM SPEAKER(S)	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: NO SOUND OUTPUT FROM SPEAKER(S)	
	1 Check speaker operation
	Is the harness connector securely connected to the audio head unit and the speaker unit(s)? Yes Proceed to the next step GO to A2. No Reconnect wiring harness to audio head unit/speaker unit(s)
A2: NO SOUND OUTPUT FROM SPEAKER(S)	
	1 Check audio head unit operation
	Is the audio head unit operational?

	Yes Proceed to the next step GO to A3. No Check the integrity of the power supply circuits/fuses to the audio head unit and rectify as required
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A3: NO SOUND OUTPUT FROM SPEAKER(S)

	1 Check which speakers are operational
	Are all speakers working? Yes Proceed to the next step GO to Pinpoint Test B. No Use the fader control to direct audio output to different speaker locations to establish which units are non-operational. Refer to the electrical circuit diagrams and check the circuits between the audio head unit and the affected speaker units for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required. If fault persists, replace non-operational speaker unit(s) as required

PINPOINT TEST B : POOR OR WEAK SOUND OUTPUT FROM SPEAKER(S)

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: POOR OR WEAK SOUND OUTPUT FROM SPEAKER(S)	
	1 Check connections to speaker units
	Are the connectors inserted securely into the speaker units? Yes Proceed to the next step GO to B2. No Ensure all connectors are securely attached to the speaker units
B2: POOR OR WEAK SOUND OUTPUT FROM SPEAKER(S)	
	1 Check integrity of vehicle power supply fuses
	Are the vehicle/audio head unit power supply fuses functional? Yes Proceed to the next step GO to B3. No Replace fuse(s) as required
B3: POOR OR WEAK SOUND OUTPUT FROM SPEAKER(S)	
	1 Check power and ground circuits to the infotainment system components
	Are all the necessary power and ground feeds present? Yes Proceed to the next step GO to B4.

	No Refer to the electrical circuit diagrams and check the infotainment power and ground circuits for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required
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B4: POOR OR WEAK SOUND OUTPUT FROM SPEAKER(S)

	1 Check power supply voltage at power supply connectors
	Is the power supply voltage measured at the power supply connectors between 12 and 14 volts? Yes No further action No Refer to the electrical circuit diagrams and check the infotainment power supply circuits for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required. Refer to the relevant section of workshop manual and battery care manual. Check battery state of charge and starting/charging system performance and rectify as required

PINPOINT TEST C : SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check for extreme bass/treble settings
	Is the audio system output settings for bass and/or treble set too high? Yes Adjust settings to appropriate levels No Proceed to the next step GO to C2.
C2: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check speaker unit(s) fixing screws are securely fastened
	Are all speaker fixing screws fully secured to the surrounding trim? Yes Proceed to the next step GO to C3. No Tighten the fixing screws to the correct torque as directed in the workshop manual
C3: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check if the wiring harness is resting against the internal surface of the speaker
	Is there any cabling or other parts of the wiring harness resting against the internal surface of the speaker? Yes

	Re-route and secure the wiring harness so that it is not resting against any internal surfaces of the speaker No Proceed to the next step GO to C4.
C4: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check security of trim, harnesses and paper labels in the vicinity of the speaker units
	Are these items secure in the vicinity of the speaker units? Yes Proceed to the next step GO to C5. No Check whether gentle contact with these items, where appropriate, relieves symptoms. If symptoms are relieved, fix source securely and prevent vibration
C5: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Remove speaker and hold in hand, taking care not to hold any moving parts of the assembly and not to damage the foam gaskets
	Does buzzing persist with speaker in hand? Yes Check for obvious debris in speaker. If there is debris and no damage, remove debris and retest for buzzing – if buzzing is fixed re-fit speaker, else replace speaker. If speaker is damaged, GO to C6. No Re-fit speaker and further investigate rattle due of nearby trim, modules, harnesses and components
C6: SPEAKER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check for an obvious source of damage to prevent any replacement also becoming damaged
	Is there an obvious source of the speaker damage? Yes Remedy source of damage and fit replacement speaker No Fit replacement speaker
C7: AUDIO OUTPUT DISTORTED	
	1 Check for extreme bass/treble settings
	Is the audio system output settings for bass and/or treble set too high? Yes Adjust settings to appropriate levels No No further action

Pinpoint Tests For Suspected Sub-Woofer Faults

NOTE:

See separate Pinpoint Tests (above) for other speaker faults

PINPOINT TEST D : NO SOUND OUTPUT FROM SUB-WOOFER	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: NO SOUND OUTPUT FROM SUB-WOOFER	
	1 Check sub-woofer operation
	Is the harness connector securely connected to the audio amplifier module and the sub-woofer unit? Yes Proceed to the next step GO to D2. No Reconnect wiring harness to audio amplifier module/sub-woofer unit
D2: NO SOUND OUTPUT FROM SUB-WOOFER	
	1 Check Integrated Audio Module (IAM) operation
	Is the integrated audio module operational? Yes Proceed to the next step GO to D3. No Check the integrity of the power supply circuits/fuses to the integrated audio module and rectify as required. Check the integrated audio module for related DTCs and refer to the relevant DTC index
D3: NO SOUND OUTPUT FROM SUB-WOOFER	
	1 Check which sub-woofers are operational
	Are all sub-woofers working? Yes Proceed to the next step GO to D4. No Use the fader control to direct audio output to different sub-woofer locations to establish which units are non-operational. Refer to the electrical circuit diagrams and check the circuits between the audio amplifier module and the affected sub-woofer units for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required. If fault persists, replace non-operational sub-woofer unit(s) as required
D4: NO SOUND OUTPUT FROM SUB-WOOFER	
	1 Check for any visible loose harness connections on sub-woofer drive unit

	Are there any visible loose harness connections (ie: loose wires / pins) on the sub-woofer Yes Replace the sub-woofer unit(s) as required No No further action
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PINPOINT TEST E : POOR OR WEAK SOUND OUTPUT FROM SUB-WOOFER(S)

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: POOR OR WEAK SOUND OUTPUT FROM SUB-WOOFER	
	1 Check connections to sub-woofer units
	Are the connectors inserted securely into the sub-woofer units? Yes Proceed to the next step GO to E2. No Ensure all connectors are securely attached to the sub-woofer units
E2: POOR OR WEAK SOUND OUTPUT FROM SUB-WOOFER	
	1 Check integrity of vehicle power supply fuses
	Are the vehicle/audio amplifier module power supply fuses functional? Yes Proceed to the next step GO to E3. No Replace fuse(s) as required
E3: POOR OR WEAK SOUND OUTPUT FROM SUB-WOOFER(S)	
	1 Check power and ground circuits to the infotainment system components
	Are all the necessary power and ground feeds present? Yes Proceed to the next step GO to E4. No Refer to the electrical circuit diagrams and check the infotainment power and ground circuits for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required
E4: POOR OR WEAK SOUND OUTPUT FROM SUB-WOOFER(S)	
	1 Check power supply voltage at power supply connectors
	Is the power supply voltage measured at the power supply connectors between 12 and 14 volts? Yes No further action No

Refer to the electrical circuit diagrams and check the infotainment power supply circuits for short circuit to ground, open circuit, high resistance. Repair circuit(s) as required. Refer to the relevant section of workshop manual and battery care manual. Check battery state of charge and starting/charging system performance and rectify as required

PINPOINT TEST F : SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check for extreme bass/treble settings
	Is the audio system output settings for bass and/or treble set too high? Yes Adjust settings to appropriate levels No Proceed to the next step GO to F2.
F2: SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check sub-woofer unit(s) fixing screws to body are securely fastened
	Are all sub-woofer fixing screws fully secured to the body ? Yes Proceed to the next step GO to F3. No Tighten the fixing screws to the correct torque as directed in the workshop manual
F3: SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY	
	1 Check security of trim, harnesses and paper labels in the vicinity of the subwoofer(s)
	Are these items secure in the vicinity of the subwoofer(s)? Yes Proceed to the next step GO to F4. No Check whether gentle contact with these items, where appropriate, relieves symptoms. If symptoms are relieved, fix source securely and prevent vibration
F4: SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY	
	1 Remove subwoofer unit(s) (do not disassemble part) and hold in hand, taking care not to hold any moving parts of the assembly and not to damage the foam gaskets
	Does buzzing persist with subwoofer unit(s) in hand? Yes Check for obvious debris in subwoofer unit(s). If there is debris and no

	damage, remove debris and retest for buzzing – if buzzing is fixed re-fit subwoofer unit(s), else replace subwoofer unit(s). If subwoofer unit(s) damaged, GO to F5. No Re-fit subwoofer unit(s) and further investigate rattle due of nearby trim, modules, harnesses and components
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F5: SUB-WOOFER VIBRATING (BUZZING) EXCESSIVELY

	1 Check for an obvious source of damage to prevent any replacement also becoming damaged
	Is there an obvious source of the subwoofer unit(s) damage? Yes Remedy source of damage and fit replacement subwoofer unit(s) No Fit replacement subwoofer unit(s)

PINPOINT TEST G : SUB-WOOFER AUDIO OUTPUT DISTORTED

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: SUB-WOOFER AUDIO OUTPUT DISTORTED	
	1 Check for extreme bass/treble settings
	Is the audio system output settings for bass and/or treble set too high? Yes Adjust settings to appropriate levels No Re-check possible sub-woofer faults GO to Pinpoint Test D.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.